Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/12/2006
PI Name:	Biswas, Gautam Ph.D.		
Project Title:	Distributed Monitoring and Control of Complex Dynamic Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	37235	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	03/10/2003	**End Date:**	03/09/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kortenkamp, David (Metrica, Inc.)		
Grant/Contract No.:	NCC9-159		
Performance Goal No.:			
Performance Goal Text:			
	The goal of this project is to develop an integrated, robust fault-adaptive control approach for complex dynamic systems by combining distributed monitoring, condition-based maintenance, and decision making to maintain system objectives and support long-term operation of the system. The model-based methodologies developed will be applied to subsystems of the Advanced Life Support (ALS) System being developed for long-duration manned missions. This proposal falls into the basic research category (levels 1 and 2), but we will evaluate the effectiveness of this methodology by developing advanced control experiments on selected Life Support System problems. The Life Support System for the International Space Station and future manned lunar and Martian exploration provides interesting challenges in that they include a number of interacting control loops, such as the energy management loop, the thermal control loop, the bio-regeneration and gas transfer loop, and the chemical production loop, that cover a variety of domains. An added challenge is that these loops operate at very different time scales, and this will require us to develop		

Task Description:	innovative abstraction mechanisms and introduce hierarchical model composition techniques to handle concurrency problems in modeling and analysis for autonomous and robust control. Task Objective: Develop multi-level fault-adaptive control schemes to manage complex distributed systems, such as the Advanced Life Support systems for long-duration manned missions. Task Significance: 1. Systematic approaches to modeling complex, distributed systems ; 2. Distributed algorithms for online monitoring and fault isolation ; 3. Multi-level fault adaptive controller design that meet resource constraints while optimizing performance over long time horizons
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Water Recovery Systems may be used in desert regions. Advanced autonomous control will make these systems more robust, efficient, and long-lasting. Modeling, monitoring, and control technologies and tools developed can be applied to the design and analysis of other complex, dynamic systems.
Task Progress:	This project focused on developing autonomous, fault-adaptive, real-time control technology for complex dynamic systems. The control approaches developed are tailored to autonomously detect and isolate faults in system operation and reconfigure the controller to ensure stable and effective system operation. Faults refer to drifts or failures in actuators, sensors, and system components. The detection and isolation of these faults are accomplished through hybrid system observers, statistical fault detection modules, fault isolation and identification units that combine qualitative reasoning with temporal causal graphs and parameter estimation with state space models. A key novelty of this project was the development of fault observers and adaptive control systems for hierarchical hybrid systems, i.e., systems containing both continuous dynamics and discrete events. A hierarchical modeling scheme efficiently deals with interacting subsystems for online supervisory control. Model-based techniques provide the overarching framework for developing the individual component technologies. The primary advantage of such systems will be the dramatic reduction of manually intensive monitoring and control. Another distinct advantage is improved safety due to the continuous fault detection and adaptation capability. Tasks Completed. 1. Unified modeling language for hybrid physical systems. Component-oriented model development environment. Ability to deveop different executable modeling forms from basic Hybrid Bond Graph representation. 2. Online monitoring and fault detection for complex hybrid systems in noisy environments. 3. Fault isolation and identiifcation technologies for hybrid systems. 4. Online, hierachical, fault-adaptive control for distributed systems that combines resource management and multi-level hybrid optimizing control approaches in a decision-theoretic framework. 5. Applications of the methodologies developed to ALS subsystems and 90 day challenge problem developed at NASA Johnson Space Center (JSC). 6. Extensions of the framework for integrated planning and control. Innovative technologies developed: 1. Graphical Hybrid Modeling Environment that allows one to build models of components, and builds the overall system model using compositional techniques. The modeling environment is linked to an interpreter that automatically generates Matlab/Simulink models for simulation studies. 2. A complete environment for fault detection and fault isolation in hybrid dynamic systems. 3. Multi-level control schemes that was recently demonstrated for Advanced Life Support systems for a four crew 90 day lunar surface mission.
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